

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

GET 202 — Past Questions & Materials
by Renance

Course: GET 202
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 200

This study pack was compiled and restructured by Renance for GET 202 students at Federal University of Petroleum Resources, Effurun. It brings together 1 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

Study online: renance.app
Get the app: Google Play — [dev.renance.renance](https://play.google.com/store/apps/details?id=dev.renance.renance)

Generated by Renance AI

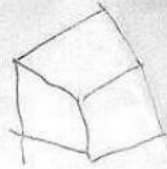
Paper 1 of 1 — GET 202. Engineering marterial 2024/2025 Past Question.

Level 200 • Second Semester • PastQuestion

This source paper is a scanned document; its pages are reproduced below.

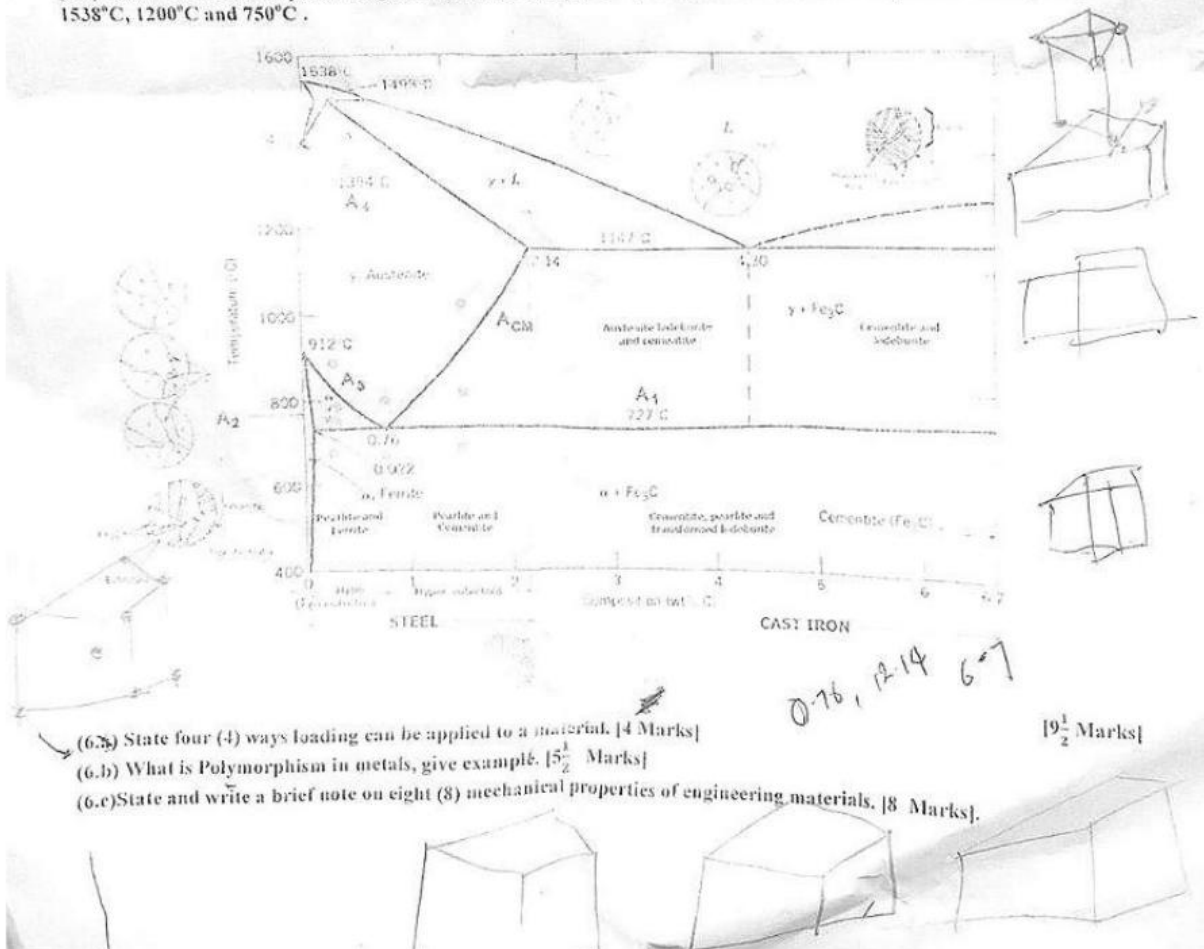


FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
DEPARTMENT OF MECHANICAL ENGINEERING
 2024/2025 Academic Session, Second Semester Examination
COURSE CODE AND TITLE: GET 202/ MEE 214 – Engineering Materials.
DATE: September 23rd, 2025. TIME ALLOWED: 2 Hours
INSTRUCTIONS: Answer four (4) questions only.



- ✓ (1. a) With the aid of a tree diagram classify engineering materials. [4 Marks]
 (1.b) State the seven crystal lattice systems and give example each of the element/compound that adopts their crystal system. [4 Marks]
 (1.c) A cylindrical alloy steel rod was prepared as a specimen for a tensile test before it was further used for the production of automobile universal shaft. If the specimen was loaded with 350N weight and subjected to a stress of 56 MPa. Calculate the diameter of the specimen. [9½ Marks]
- ✓ (2.a) What is heat treatment. [4 Marks]
 (2.b) State seven (7) objectives of heat treatment. [4 Marks]
 (2.c) List and write a brief note on five (5) heat treatment processes commonly employed in engineering practice. [9½ Marks]
- ✓ (3.a) With the aid of a neat sketch draw the three crystal lattice usually found in most of the metals. [6 Marks]
 (3.b) The unit cell of chromium is cubic and contains 5 atoms. Calculate the lattice constant in terms of the radius. Given that Cr = 52g, Avogadro Number (AN) = 6.02×10^{23} . Density of chromium = 7.2 Mg/m^3 . [9½ Marks]
 (4.a) A binary alloy consisting of 40% A and 60% B, and at a certain temperature, have two phases, α (with 30% B) and β (with 70% B), in equilibrium. Determine the fractions of each phase. [9½ Marks]
 (4.b) State the three main types of binary phase diagram.
 (4.c) State some applications of Phase diagram. [4 Marks].
- ✓ (5.a) Discuss the effect of impurities on cast iron. [8 Marks]
 (5.b) Below is the Fe-C phase diagram, calculate the phases present at the eutectoid composition line at, temperature of 1538°C, 1200°C and 750°C.

hemange



- ✓ (6.a) State four (4) ways loading can be applied to a material. [4 Marks]
 (6.b) What is Polymorphism in metals, give example. [5½ Marks]
 (6.c) State and write a brief note on eight (8) mechanical properties of engineering materials. [8 Marks].

